

Gym System

Business Problem :

- Paper-Based
 - only offline registration
 - No attendance tracking
 - No subscription expiry alerts
 - No trainer–member communication
 - No class scheduling system
 - Overcrowding at peak hours
 - No performance tracking (weight / body)
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Vision :

- Centralized system
- Controlled Classes
- Tracable funds
- **Member awareness**
- Users Role
- attendance & performace tracking
- Accurate invoicing

Product Requirements Document (PRD)

Product: Gym Management System

Version: 1.0

Date: <25/4 /2026>

1. Product Overview

The Gym Management System is a centralized platform designed to manage gym members, subscriptions, payments, attendance, classes, and performance tracking.

The system replaces manual and paper-based processes with a structured digital solution, improving operational efficiency, financial tracking, and member experience.

2. Problem Statement

Gyms currently face several operational challenges:

- Reliance on paper-based systems
- No centralized tracking for members and subscriptions
- No attendance tracking
- No visibility into payments and revenue
- Lack of communication between coaches and members
- No class scheduling or capacity control
- Overcrowding during peak hours
- No tracking of member progress (weight, performance)

This system aims to provide a **single source of truth** for gym operations.

3. Goals & Objectives

- Digitize all gym operations
- Improve tracking of memberships and payments

- Provide visibility into member activity and attendance
 - Enable structured class scheduling and capacity control
 - Improve communication between coaches and members
 - Enhance member awareness (subscription, progress, workouts)
 - Reduce operational errors and manual workload
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4. User Roles & Permissions

Role	Permissions
Admin	Full access: manage members, subscriptions, payments, classes, reports
Coach	View assigned members, assign workouts, track performance, manage sessions
Member	View subscription, track progress, check-in, book classes

5. Functional Requirements

5.1 Member Management

As an Admin, I need to:

- Register new members
- Update member details
- Deactivate members
- View member status

Acceptance Criteria:

- Member must have unique identifier
 - Cannot delete member with active subscription
 - All required fields must be validated
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5.2 Subscription Management

As an Admin, I need to:

- Create subscription for a member
- Assign a plan (with price snapshot)
- Activate subscription after successful payment
- View subscription status (Active / Expired)

Business Rules:

- Subscription cannot be active without successful payment
 - Subscription stores price snapshot (not dynamic plan price)
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5.3 Payment Management

As an Admin, I need to:

- Record payments
- View payment history
- Track successful and failed payments

Business Rules:

- Payment must be linked to a subscription
 - Payment amount must match subscription price
 - Subscription activates only after successful payment
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5.4 Attendance Tracking

As a Member/Admin, I need to:

- Check-in members
- Track attendance history

Business Rules:

- Member cannot check-in without active subscription
 - Attendance is recorded with timestamp
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5.5 Class Management

As an Admin, I need to:

- Create classes with schedule and capacity
- Assign coach to class
- Manage class availability

As a Member, I need to:

- Book a class

Business Rules:

- Cannot exceed class capacity
 - Cannot double-book member in overlapping classes
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5.6 Performance Tracking

As a Coach, I need to:

- Track member progress (weight, performance)
- Update workout plans

Acceptance Criteria:

- Progress entries must be timestamped
 - Coaches can only access assigned members
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5.7 Dashboard & Reporting

As an Admin, I need to:

- View total members

- View active vs expired subscriptions
 - Track revenue
 - View attendance statistics
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5.8 Authentication & Authorization

- Users must log in
 - Role-based access control enforced
 - Secure password storage
 - Session management
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6. Non-Functional Requirements

Performance

- API response < 500ms
- Supports concurrent users

Security

- Encrypted data (HTTPS)
- Role-based access
- Session expiration

Usability

- Simple and intuitive UI
- Clear validation messages

Data Integrity

- Cannot activate subscription without payment
 - Referential integrity maintained
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7. User Stories

Story 1: Create Subscription

- As an Admin
 - I want to create a subscription and record payment
 - So that the member can access the gym
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Story 2: Member Check-in

- As a Member
 - I want to check-in to the gym
 - So that my attendance is recorded
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Story 3: Book Class

- As a Member
 - I want to book a class
 - So that I can attend sessions
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8. Out of Scope

- Multi-branch gyms
 - Mobile application
 - Advanced analytics
 - External integrations
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9. Success Metrics

- 100% subscription tracking accuracy
- Reduced manual errors

- Increased member retention
 - Improved revenue visibility
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10. Assumptions & Dependencies

- Single gym location
 - Limited number of users
 - Internet availability
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11. Constraints

- Web-based system only
 - No multi-tenancy
 - English language
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